

Handwritten Text Recognition

Part I Acquisition and preprocessing of text images

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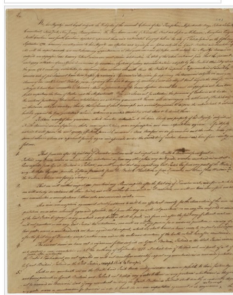
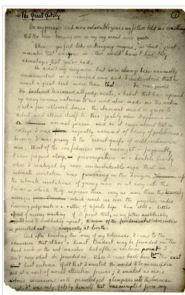
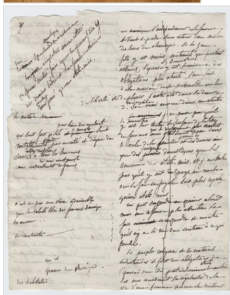
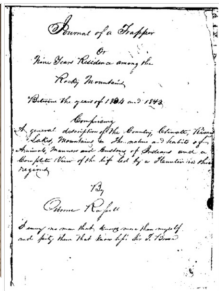
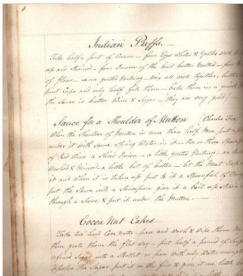
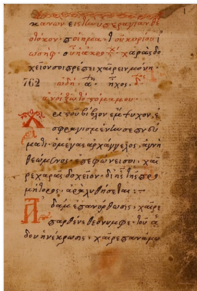
Outline

- Introduction: HTR Generalities
 - Motivation
- HTR System Overview
- Text Images Acquisition
- Document Image Processing
 - Noise Reduction
 - Background Removal
 - Warping Correction
 - Skew Correction
 - Text Line Image Detection and Extraction
- Text Line Image Processing
 - Slope Correction
 - Slant Correction
 - Size Normalization
 - Characters Width
 - Stroke Thickness

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Text pages examples



Writing styles

PRECISION EN LAS
MODIFICACIONES SOLICI-
TADAS, EN EL 609 CAL.

* INFORMACIÓN SOBRE "BUENOS"
PRECIOS DE LAS TERMINALES
→ MANTENIENDO EL NÚMERO →
* MÁS INFORMACIÓN SOBRE EL
PROGRAMA DE PUNTOS

Yo comprar el móvil, para
usarlo en mi departamento.
dijo GERARTE VILLA, (ahora
y se me dijo que había "cobertura" para mí
y para hablar hay que salirse no sólo de
la casa. sino del banco del pueblo..

que tus facturas
lleguen antes del
corte en BANCO

Writing styles

malfuncionamiento desservicios

UTILIZAMOS EL SERVICIO

HESTOI

~~HESTOI~~ AVECES LA

el establecimiento de Hacienda

DEBERIAN DAR MÁS FACILIDAD

Maya Inglaes para cambiar
de telefono instalando el mismo
N.º cuando sea por accion
a rotura por roturas del
apunto. (Me refiero al precio)

POR EL MOMENTO ~~HESTOI~~ QUE ESTOY

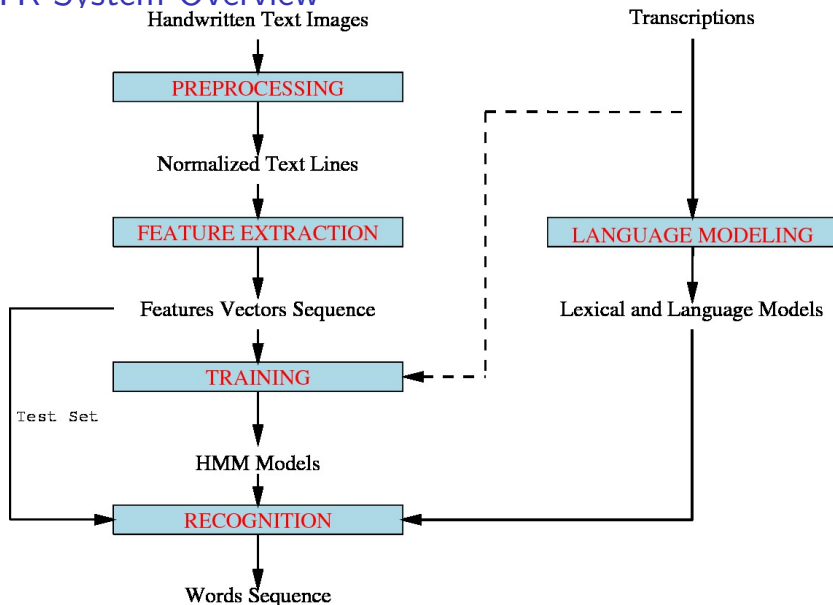
Motivation

- ▶ At a page level, geometric normalization aim to help the layout analysis and the line extraction process
- ▶ At a line level, the large variability on the handwriting style do not help the recognition process
- ▶ The aim is to uniformise as possible the writing style

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HTR System Overview



Outline

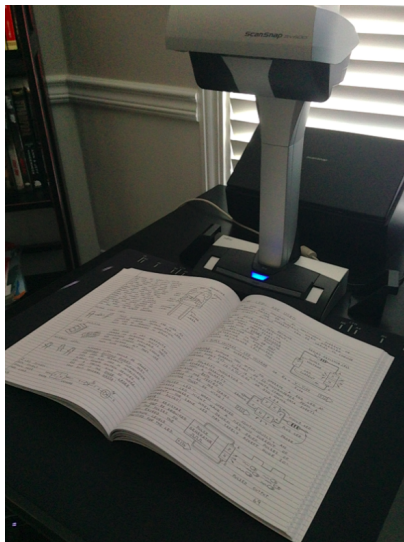
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Text Images Acquisition

Flatbed Scanner A light source shines on the image. A series of mirrors redirects the light reflected from the original image to a focusing lens. This lengthens the effective focal length of the lens. The light is then directed to the sensor array, which collects charge as the bar moves continuously along the page.

Cameras and Mobile Devices similar to those of flatbed scanners, but the position of the sensor relative to the object is now variable and so is the illumination. This increases the range of image distortions.

Scanning Books



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Document Image Processing

Preprocessing steps commonly applied to handwritten text pages images:

Noise Reduction: filtering out noise, which can be due to an intrinsic feature of the original image and/or the digitization process.

Background Removal: aiming at removing everything that is not handwritten text.

Binarization: transforming (mapping) scaled grey level images into a black-white images.

Warping Correction: correction of the curl distortions due to the scanning of non planar surfaces as ie. books.

Skew Correction: detection and correction of the page angle with respect to the horizontal direction.

Layout analysis: identifying and categorizing the regions in a scanned image of a text document.

Text Lines Extraction: detection and extraction of text lines from page image.

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Noise Reduction

Issues;

- ▶ Stains, smears, ink bleed through.
- ▶ Uneven illumination. contrast variation.
- ▶ Noisy background.
- ▶ Salt-and-pepper.
- ▶ etc.

Solution:

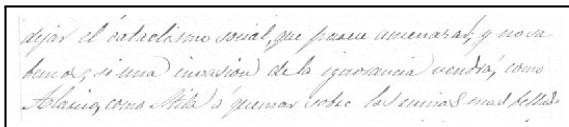
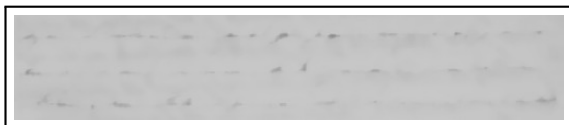
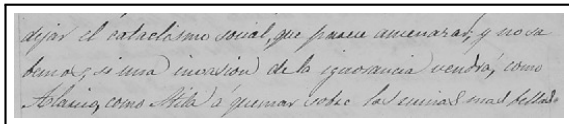
Methods based mainly on (combination of) different thresholding approaches, connected components and image Preprocessing filters.

Popular thresholding methods:

- ▶ N. Otsu, "A threshold selection method from grey level histogram", IEEE Trans. Syst. Man Cybern., vol. 9 no. 1, 1979, pp. 62-66. Global thresholding method.
- ▶ W. Niblack, "An Introduction to Digital Image Processing", pp. 115-116, Prentice Hall, 1986. Local thresholding method.
- ▶ J. Sauvola, M. Pietikäinen, "Adaptive document image binarization", Pattern Recognition, Volume 33, Issue 2, February 2000, Pages 225-236, ISSN 0031-3203. Local thresholding method.

Background Removal

Taking out low frequencies: **median filter**.



Binarization

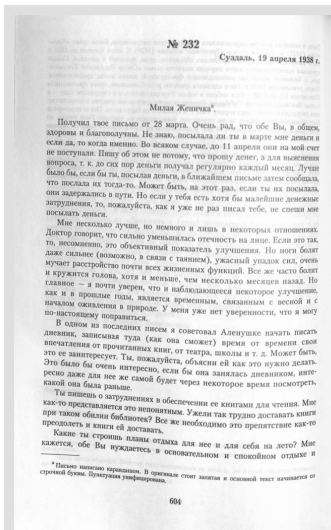
“Important and critical stage in the document image analysis and recognition pipeline since it permits less image storage space, enhances the readability of text areas and allows efficient and quick further processing for page segmentation and recognition” (Basilis Gatos)

Binarization methods:

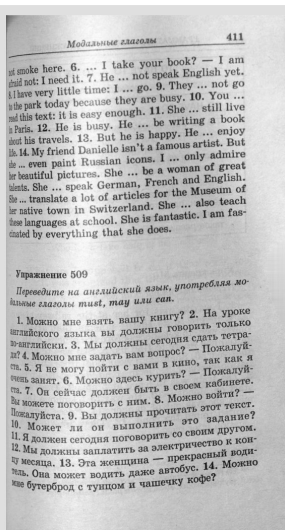
Handbook of Document Image Processing and Recognition, Doermann, David; Tombre, Karl (Eds.) ISBN 978-0-85729-858-4 June 30, 2014

Reference	Category	Short description	Remarks
Otsu [3] (1979)	Global thresholding	The optimum threshold is calculated by separating the two classes of pixels (e.g. foreground and background) so that their between-class variance is maximized.	Successful only to document images having bimodal distributed histogram.
Chariet et al. [4] (1998)	Global thresholding	A recursive application of [3] in order to treat document images with multiple background.	Demonstrated for the binarization of bank checks.
Solihin et al. [5] (1999)	Global thresholding	A two-stage thresholding approach requiring at a first stage each pixel to be assigned to one of three classes (foreground, background, fuzzy area).	Designed to binarize gray-scale handwriting images.
Niblack [6] (1986)	Local thresholding	A local binarization algorithm that calculates a pixelwise threshold by shifting a rectangular window across the image.	Noise that is present in the background may remain dominant in the final binary image.
Sauvola and Pietikainen [7] (2000)	Local thresholding	A modification of [6] that adds a hypothesis on the gray values of text and background pixels.	Noise is eliminated but text regions may be missed.
Kamel and Zhao [8] (1993)	Local thresholding	The character stroke width and a global threshold are used in order to detect significant changes in gray level values around the character body.	It is very difficult to tune the parameters of the method.
Yang and Yan [9] (2000)	Local thresholding	A modification of [8] in which character stroke width is automatically detected and the threshold is locally adapted.	For poor quality gray-scale document images without need of any manual parameter tuning.
Gatos et al. [10] (2006)	Local thresholding	Adaptive method based on pre-processing, background surface calculation, suitable thresholding and post-processing.	For degraded documents with shadows, non-uniform illumination, low contrast and noise.
Kim et al. [11] (2002)	Hybrid thresholding	Global thresholding [3] on a difference image between the original and the water-filled image.	Selection of two critical parameters on an experimental basis.
Vonikakis et al. [12] (2011)	Hybrid thresholding	It combines the characteristics of the OFF-center-narrowed cells of the Human Visual System with the classic Otsu binarization technique [3].	Experiments performed on a set of images with various computer-generated degradations.
Tsang and Lee [13] (2008)	Hybrid thresholding	Based on analyzing the document layout and the intensity distribution. Pixels outside the detected blocks are binarized using Otsu technique [3].	Tested on a large set of images including newspapers, magazine articles and business cards.
Beddoles and Papamarkos [14] (2007)	Combination of techniques	Combines the results of several global and local binarization algorithms using a Kohonen self-organizing map neural network.	Tested on a variety of degraded document images.
Gatos et al. [15] (2008)	Combination of techniques	Combines the binarization results of several global and adaptive methodologies using a majority voting strategy and incorporates edge map information.	For historical and degraded document images.
Su et al. [16] (2011)	Combination of techniques	Image pixels are labeled as foreground, background or uncertain based on the results of several binarization techniques. Uncertain pixels are classified at a next step as foreground or background.	It improves the performance of existing binarization methods, robust on degraded images.
Chou et al. [17] (2010)	Using training samples	Divides the image into regions and decides how to binarize each region using a learning process and a support vector machine (SVM) classifier.	Better visual quality and OCR performance than several global and local methods.
Huang et al. [18] (2008)	Using training samples	Hidden Markov model (HMM) based binarization using feature vectors calculated from several image directions.	Better OCR performance than other conventional methods.
Chen and Luedham [19] (2005)	Using training samples	Local feature vectors are used to analyze and find the best approach to threshold a local area after quad-tree decomposition.	Trained using 300 historical images and tested on 300 degraded document images.
Beddoles et al. [20] (2006)	For color documents	A color reduction procedure is followed by a connected component analysis in order to classify components as text or non-text.	Appropriate for noisy color or grayscale documents.
Tsai and Lee [21] (2002)	For color documents	The color image is transformed into the luminance and saturation spaces. A decision tree is used to decide if luminance, saturation or both features will be employed for the binarization process.	Trained on 150 different color documents and tested on an additional 319 color documents.
Wang et al. [22] (2005)	For color documents	Color text image binarization based on color clustering and binary texture feature analysis. Two kinds of texture features, run-length histogram and spatial-size distribution related respectively, are extracted and explored.	Experiments with more than 500 color text images collected from the Internet.

Warping pages example



* Письмо написано карандашом. В оригинале стоит заглавия и основной текст начинается со строчной буквы. Пунктуация унифицирована.



Dewarping approaches

- ▶ Directly at restoring straight text lines. Zhang Z, Tan CL (2001) "Restoration of images scanned from thick bound documents". In: Proc. ICIP, pp 1074-1077.
- ▶ Fitting a quadratic polynomial curve to the distorted text line. Zhang Z, Tan CL (2003) "Correcting document image warping based on regression of curved text lines". In: Proc. ICDAR, pp 589-593.
- ▶ Modelling the 3D shape of the page. Pitu M (2001) "Undoing paper curl distortion using applicable surfaces". In: Proc. CVPR, pp 67-72.

Skew Correction of Page Image

Issues:

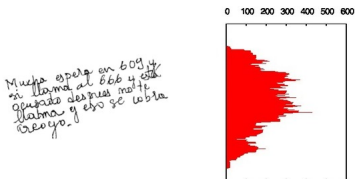
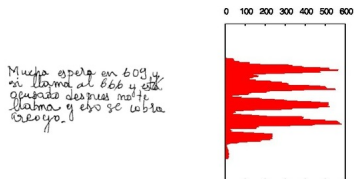
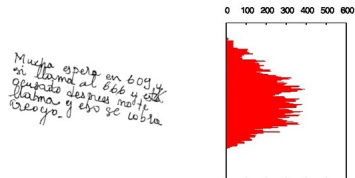
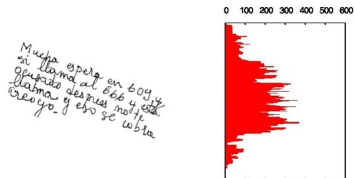
- ▶ Introduced during the image scanning process (or due to the writer).
- ▶ Defined as the angle between the X-axis and the scanned text lines of the page.
- ▶ Solution: rotation transform when the skew angle is detected.
- ▶ Its correction benefits the Line Extraction process.

Typical methods based on:

- ▶ Multiresolution, Eigenvalues, Connected components
- ▶ Hough transform
- ▶ Horizontal projection profiles
- ▶ Statistical mixture model (EM algorithm)
- ▶ etc.

B. Gatos, N. Papamarkos, and C. Chamzas. Skew detection and text line position determination in digitized documents. *Pattern Recognition*, 30(9):1505-1519, 1997.

Skew: projections



Skew: projections

- ▶ A projection can be calculated as

$$\text{HPP}_{\alpha}(i) = \sum_{j=1}^{\text{cols}} I_{\alpha}(i, j)$$

- ▶ The problem can be reduced to an optimization one

$$\hat{\alpha} = \arg \max_{\alpha \in [\alpha_{\min}; \alpha_{\max}]} \mathbf{F}(\text{HPP}_{\alpha})$$

- ▶ It is needed to found an good objective function
- ▶ One of the best objective function is the standard deviation

Text Line Images Detection and Extraction

There are two different issues with respect to the text lines in a given page image: **Detection** and **Extraction**.

Typical methods for text line detection are based on:

- ▶ Multiresolution, Eigenvalues
- ▶ Hough transform
- ▶ Horizontal projection profiles (+ RLSA)
- ▶ Statistical approaches: HMMs
- ▶ etc.

Typical methods for text line extraction are based on:

- ▶ Connected components
- ▶ Dynamic programming, Distance Maps (cost functions)
- ▶ etc.

Laurence Likforman-Sulem, Abderrazak Zahour, and Bruno Taconet. 2007. Text Line Segmentation of Historical Documents: A Survey. Int. J. Doc. Anal. Recognit. 9, 2 (April 2007), 123-138.

Baseline detection: Introduction

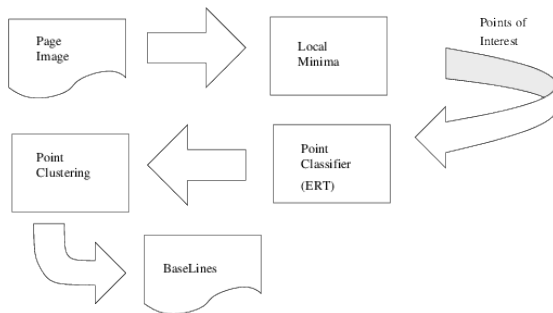


- ▶ There are loads of manuscript scanned page images.
- ▶ HTR or Catti systems need to be fitted with images containing just a line.
- ▶ Segmenting pages into lines images is a really expensive process.
- ▶ *Bottle-neck*: holding back the massive industrial document processing.

Baseline detection: introduction

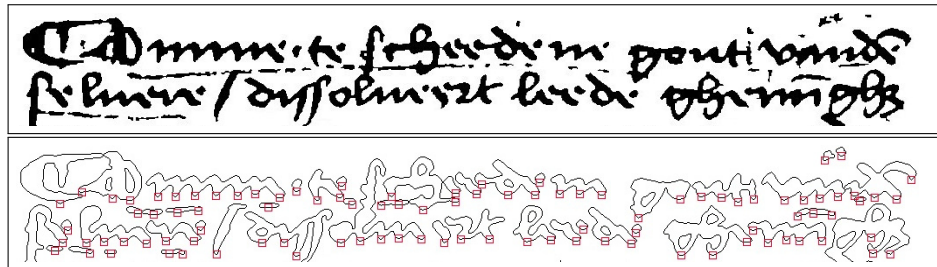
- ▶ *Baseline*: fictitious poly-line which follows the lower part of the character bodies.
- ▶ *Baseline detection*: a process to know where the text lines are.
- ▶ Not only to segment page images, but also for other processing steps: skew, slant, slope correction, text height normalization or feature extraction.
- ▶ Not an easy process in manuscript texts: different skews, variable inter-line spacing, overlapping and touching words between adjacent lines, etc.
- ▶ Even worse in historical documents due to the degradation problems: smear, significant background variations and uneven illumination, spots due to humidity, bleed through, layout inconsistencies, decorative entities, etc

Baseline Detection Algorithm



- ▶ Baseline detecting problem formulated as a clustering problem over a set of interest points.
- ▶ The method proposed here is designed with the aim to be robust to the noise artifacts that usually appears in historical manuscripts.
- ▶ Moisés Pastor, "Text baseline detection, a single page trained system", Pattern Recognition, Vol. 94, 2019, Pages 149-161, ISSN 0031-3203, <https://doi.org/10.1016/j.patcog.2019.05.031>.

Interest Points



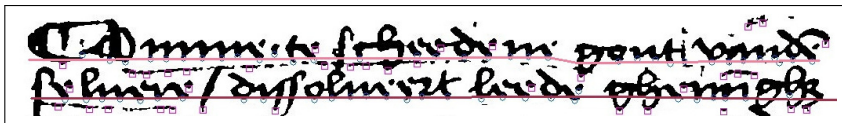
- ▶ Interest points are those points susceptible to belong to baselines.
- ▶ Local minima points, obtained from the text edges are chosen as interest points.
- ▶ After a blur and a binarization (Otsu algorithm), the contours of the foreground of the text image are obtained .
- ▶ Then, using an analysis window centered on each point of the contours, the local minima are obtained.

Extremely Randomized Trees (ERT)

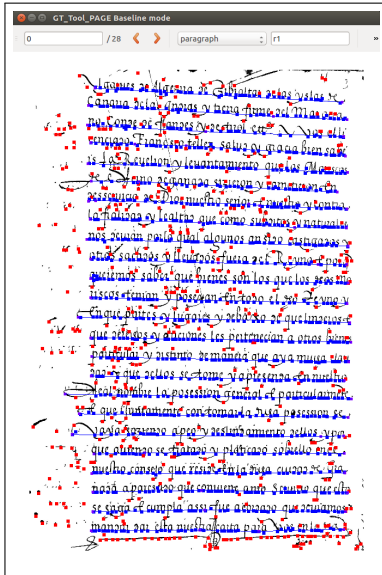
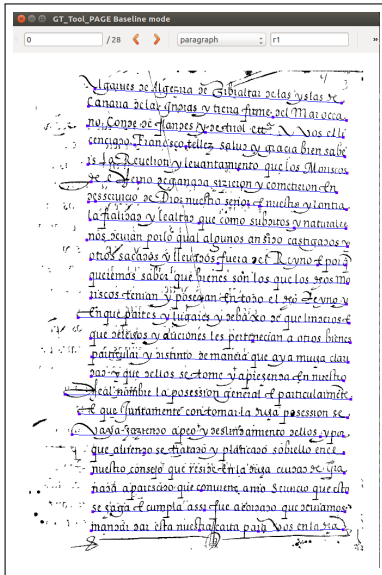
- ▶ Local minima points are usually obtained from noisily images.
- ▶ An automatic point classifier is needed to bring robustness to the noise, and other kind of artifacts
- ▶ ERT's combines the bagging idea and the random selection of features in order to get a low variance.
- ▶ ERT is fast to estimate and classifying and achieve great performance with few labeled samples to train.

Ground Truth: semi-automatic

- ▶ To train the ERT's a set of labeled points is needed.
- ▶ From the manual baselines
 - ▶ points which are close enough to are classified as belonging to (blue circles)
 - ▶ and those too much far are classified as not belonging (red squares).

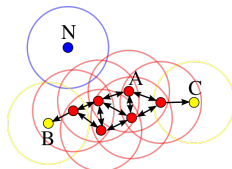


Ground Truth: manual



Clustering points into baselines: Modified DBSCAN

- DBSCAN is a density based clustering algorithm



- DBSCAN metrics were changed to use the Mahalanobis distance instead.
- This is done in order to take profit of the a-priori knowledge of the quasi-horizontal distribution of the local minima points of the text lines.

$$d_m(\vec{p}_1, \vec{p}_2) = \sqrt{(\vec{p}_1 - \vec{p}_2)^T \Sigma^{-1} (\vec{p}_1 - \vec{p}_2)} \quad (1)$$

where:

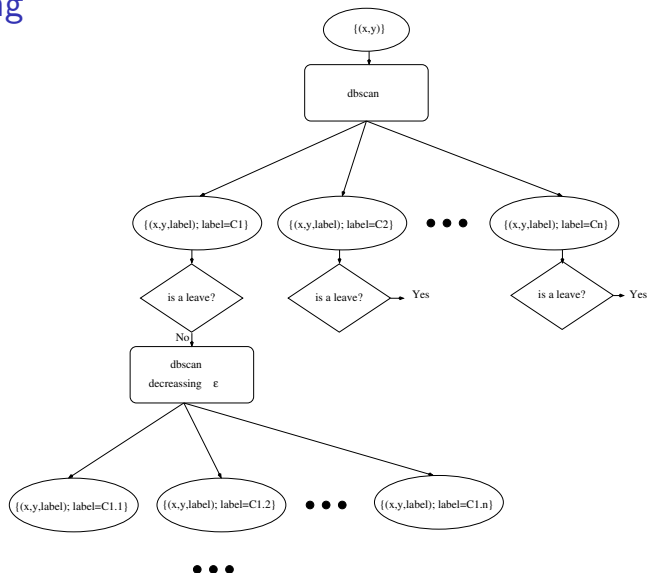
$$\Sigma = \begin{Bmatrix} \varepsilon_x & 0 \\ 0 & \varepsilon_y \end{Bmatrix}$$

And $\varepsilon_x = \varepsilon$ and $\varepsilon_y = \lambda \varepsilon$

Clustering

- ▶ A initial first partition is obtained using the DBSCAN algorithm.
- ▶ A criteria function is used to decide if each class is or not a leave, in which case it will need to be splitted.
- ▶ The criteria function is based on the analysis of the slopes of the straight lines joining two consecutive points, once those points are ordered along the x -axis
- ▶ Every time a class is proposed to be splitted, the DBSCAN neighbours area, controlled by ε is reduced and the algorithm is applied to this cluster points set.

Clustering



Clustering

George Washington was one of the Founding Fathers of the United States, serving as the commander in chief of the Continental Army during the American Revolutionary War. He also presided over the convention that drafted the Constitution, which replaced the Articles of Confederation. The Constitution established the position of President of the republic, which Washington was the first to hold. Washington was elected President as the unanimous choice of the Electoral College in 1788 and he served two terms in office.

He oversaw the creation of a strong well-financed national government that maintained neutrality in the wars raging in Europe, suppressed rebellion and won acceptance among Americans of all types. His leadership style established many forms and rituals of government that have been used since, such as using a cabinet system and delivering an inaugural address. Further, the peaceful transition from his presidency to the presidency of John Adams established a tradition that continues into the

21st century. Historically, Washington has been widely regarded as the Father of his country.

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He oversaw the creation of a strong well-financed national government that maintained neutrality in the wars raging in Europe, suppressed rebellion and won acceptance among Americans of all types. His leadership style established many forms and rituals of government that have been used since, such as using a cabinet system and delivering an inaugural address. Further, the peaceful transition from his presidency to the presidency of John Adams established a tradition that continues into the

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Outline

- Introduction: HTR Generalities
 - Motivation
- HTR System Overview
- Text Images Acquisition
- Document Image Processing
 - Noise Reduction
 - Background Removal
 - Warping Correction
 - Skew Correction
 - Text Line Image Detection and Extraction
- Text Line Image Processing
 - Slope Correction
 - Slant Correction
 - Size Normalization
 - Characters Width
 - Stroke Thickness

Text Line Image processing

Style attributes of handwritten text to be corrected/normalized:

Slope Angle: angle of the handwritten text line with respect to the horizontal direction.

Slant Angle: angle of the handwritten text strokes with respect to the vertical direction.

Height of the handwritten text strokes: which can vary according to the task and writer. Of particular interest is the relationship among the sizes of ascender letters (e.g.: b, l, t), descender letter (e.g.: p, q, j) and normal letters (e.g.: a, c, u).

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Slope and Slant Correction

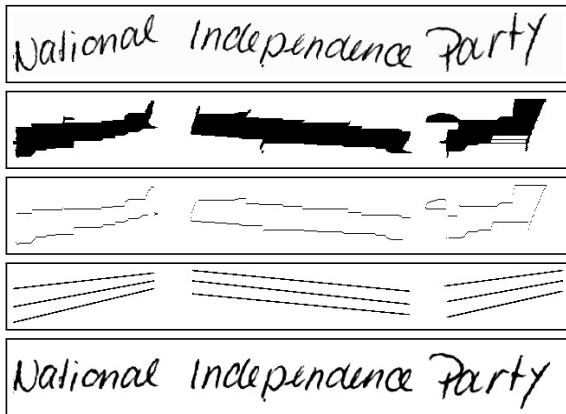
Rotation/shift transform when the slope/slant angle is detected.

Methods based on:

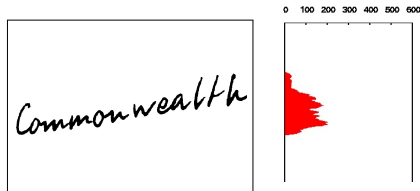
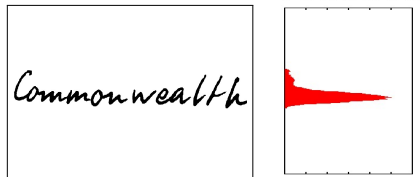
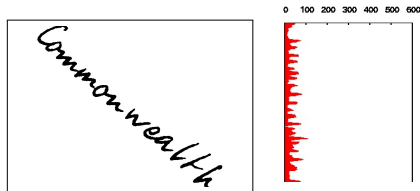
- ▶ Dynamic Programming: Bertolami, R.; Uchida, S.; Zimmermann, M.; Bunke, H., "Non-Uniform Slant Correction for Handwritten Text Line Recognition", Document Analysis and Recognition, 2007. ICDAR 2007. Ninth International Conference on , vol.1, no., pp.18,22, 23-26 Sept. 2007.
- ▶ Gabor filters: Das Gupta, J.; Chanda, B., "Novel methods for slope and slant correction of off-line handwritten text word", Emerging Applications of Information Technology (EAIT), 2012 Third International Conference on , vol., no., pp.295-298, Nov. 30. 2012-Dec. 1 2012.
- ▶ Edges detectors (Sobel filters)
- ▶ RLSA, projection profiles, linear regression (more or less heuristics)

Slope Correction: RLSA and Linear Regression

Approach based on: "Run-length Smearing Algorithm" and word slope angles computed by linear regression technique.



Slope Correction: Horizontal Projection Profile (HPP)



$$\text{HPP}_{\alpha}(i) = \sum_{j=1}^{\text{cols}} I_{\alpha}(i, j)$$

$$\hat{\alpha} = \arg \max_{\alpha \in [45:135]} \mathbf{F}(\text{HPP}_{\alpha})$$

$$\mathbf{F}(\text{HPP}_{\alpha}) = \text{std}(\text{HPP}_{\alpha})$$

$$\mathbf{F}(\text{HPP}_{\alpha}) = \max(\text{HPP}_{\alpha})$$

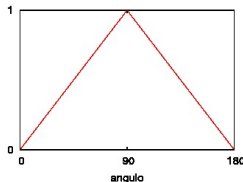
Slant Correction: Based on Edge Detection

- Derivative images produced through convolution with “Sobel” filters:

$$K_{S_h} = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix} \quad K_{S_v} = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

- Angle $\alpha_{i,j}$ is computed in each point as: $\alpha_{i,j} = \arctan \frac{(K_{S_h} * I)(i,j)}{(K_{S_v} * I)(i,j)}$
- Frequency histogram of angles is built: $H_\alpha = \sum_{\substack{i,j: \\ \alpha = \alpha_{i,j}}} 1$.
- As **slant angle**, the average value of the convolution between H_α and function F_α , is considered: $\bar{\alpha} = \sum_{\alpha=0}^{180} (H * F)_\alpha \cdot \alpha$, where:

$$F_\alpha = \left(1 - \frac{|90 - \alpha|}{90}\right)$$



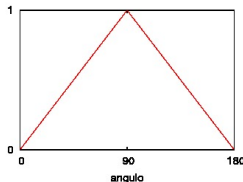
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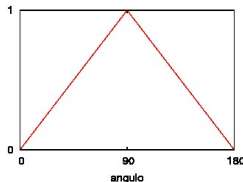
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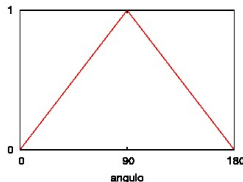
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Slant Correction: Vertical Projection Profile (VPP) (1)

- ▶ Shear transformation is applied on the image I for a wide range of angles $\alpha \in [45 : 135]$. I_α is a resulting image sheared by the angle α .
- ▶ For each sheared image the corresponding **vertical projection profile** is obtained: $VPP_\alpha(j) = \sum_{i=1}^{rows} I_\alpha(i, j)$
- ▶ The **slant angle** $\hat{\alpha}$ is that which maximizes:

$$\hat{\alpha} = \arg \max_{\alpha \in [45:135]} F(VPP_\alpha)$$

where F can be one of the following objective functions:

- ▶ Standard Deviation: $F(VPP_\alpha) = \sqrt{\sum_{j=1}^{cols} \frac{(VPP_\alpha - VPP_\alpha(j))^2}{cols}}$
- ▶ Profile Length: $F(VPP_\alpha) = \sum_{j=1}^{cols-1} \sqrt{1 + (VPP_\alpha(j) - VPP_\alpha(j+1))^2}$
- ▶ IDIAP: $F(VPP_\alpha) = \sum_{j:C_\alpha(j)=1} VPP_\alpha(j)^2$: $C_\alpha(j) = \frac{VPP_\alpha(j)}{\max(i|I(i,j)=1) - \min(i|I(i,j)=1)}$

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Slant Correction: Vertical Projection Profile (VPP) (1)

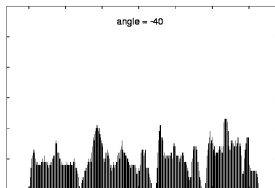
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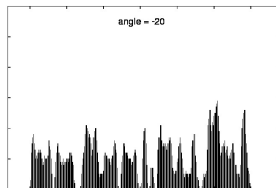
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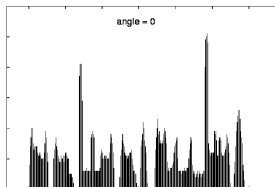
Slant Correction: Vertical Projection Profile (VPP) (2)



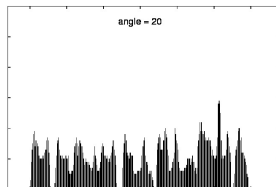
ochocientos



ochocientos



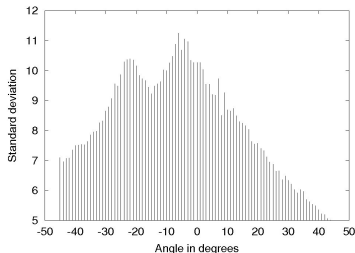
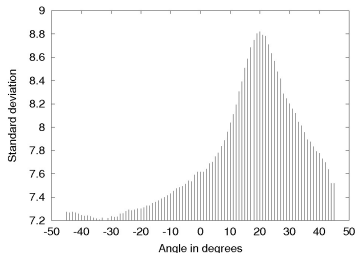
ochocientos



ochocientos

Slant Correction: Vertical Proj. Profile (VPP) (3)

When the standard deviation of vertical projection profile histogram $\mathbf{std}(\mathbf{VPP}_\alpha)$, presents a multimodal distribution:



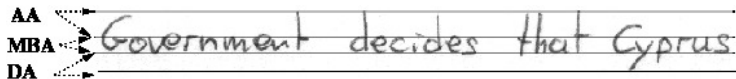
In these cases, a slant angle $\bar{\alpha}$ is computed as the average value of the angles α whose $\mathbf{std}(\mathbf{VPP}_\alpha) \geq \hat{\alpha}(1 - \rho)$:

$$\bar{\alpha} = \frac{\sum_{\alpha: \alpha \geq \hat{\alpha}(1-\rho)} \alpha \cdot \mathbf{std}(\mathbf{VPP}_\alpha)}{\sum_{\alpha: \alpha \geq \hat{\alpha}(1-\rho)} \mathbf{std}(\mathbf{VPP}_\alpha)} \quad \rho \in [0, 1]$$

Size Normalization

Encompasses two different steps: **text region detection** and **normalization**.

Into a text line image, there different areas are envisioned:



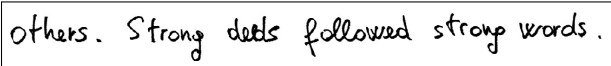
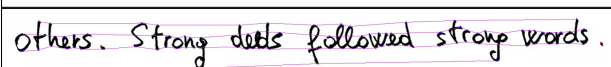
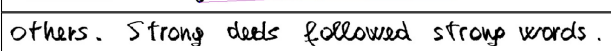
Text line normalization approaches based on:

- ▶ Conventional techniques: U.-V. Marti and H. Bunke. Using a statistical language model to improve the performance of an HMM-based cursive handwriting recognition system. International Journal of Pattern Recognition and Artificial Intelligence, 15:65-90, 2001.
- ▶ J. Gorbe-Moya, S. España-Boquera, F. Zamora-Martínez, and M.J. Castro-Bleda, "Handwritten text normalization by using local extrema classification", Pattern Recognition in Information Systems, PRIS 2008, Barcelona, pp. 164-172 Spain, June 2008.

Size Normalization: Based on RLSA and Linear Regression

1		Orig. Image
2		RLSA
3		Upper-Border
4		Lower-Border
5		Linear-Rgrs.
6		Resizing

Size Normalization: Based on Local Extrema Classification

1		Orig. Image
2		Contours
3		Local classif. points
4		Resizing